

SCIENTIFIC AMERICAN

Graphics style guide

This document provides general guidelines for creating information graphics for Scientific American online news and opinion articles. The examples within are actual published graphics showing some of the most commonly used data visualization types that appear in SciAm. Guidelines for illustrated explainers are not included, as those are much more variable in terms of visual content and illustration style (particularly when commissioned instead of produced in-house).

The style guide focuses on specifications for desktop and mobile versions of graphics and also includes a section on adapting graphics for promotional use in newsletters or social media.

Important notes about this style guide:

- Notes in magenta may signal universally applicable guidelines or may highlight where elements are styled differently for desktop and mobile. Unless otherwise specified, notes on one version of a graphic also apply to the other version of the same graphic.
- Notes in magenta pointing to graphic elements that are repeated in multiple examples (such as x- and y-axes) are only included on the first instance of each element. In all subsequent instances, the proper style is illustrated but not explicitly described using notes in magenta.
- All text follows SciAm Copy guidelines (capitalization, italics, etc.). For further guidance on issues such as abbreviations, punctuation, etc., please consult SciAm Copy Editors.
- These are all general guidelines and are subject to exceptions when necessary. If a need arises to break from these guidelines, please seek approval from SciAm Graphics Editors.

Headline
GT America Bold,
24 pts; Line height: 28

Headline for News Graphic on Desktop: Default Width

Graphic width
900 px

Intro text
GT America Regular
16 pts; Line height: 20

Intro text lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis. Aliquam convallis felis vitae metus tempus, non iaculis odio tristique.

Subhed 1
GT America
Condensed Bold,
20 pts; Line height: 23

Longer/Level 1 Subheds

Subhed 2
GT America Cond.
Bold, 16 pts, All caps;
Line height: 19

SHORTER/LEVEL 2 SUBHEDS

Label text
GT America Cond.
Reg., 16 pts;
Line height: 19

Labels look like this

Leader line
0.6 px wide, Solid dot
arrowhead at 70% scale

Annotation option 1 felis ut mollis dignissim, turpis purus ultricies libero, id bibendum ipsum nisl vitae metus

Annotation text 1
GT America Cond.
Reg., 16 pts;
Line height: 19

Annotation option 2 felis ut mollis dignissim, turpis purus ultricies libero, id bibendum ipsum nisl vitae metus

Annotation text 2
Same as 1 but 70% K
(#6D6E71)

**Footnote sodales, ex sit amet hendrerit placerat, metus lectus dapibus turpis, at tempor urna felis quis turpis.*

Footnote text
GT America Cond.
Reg. Italic, 14.5 pts;
Line height: 17;
60% K (#808285)

Graphic width
700 px

Headline for News Graphic on Desktop: Text Width

Intro text lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis. Aliquam convallis felis vitae metus tempus, non iaculis odio tristique.

Graphic width
1200 px

Headline
Align center

Headline for News Graphic on Desktop: Full Width

[Avoid intro text for full width graphics]

Headline

GT America Bold,
16 pts; Line height: 19

Headline for News Graphic on Mobile

Intro text lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis. Aliquam convallis felis vitae metus tempus, non iaculis odio tristique.

Intro text

GT America Regular,
12 pts; Line height: 16

Subhed 1

GT America Cond.
Bold, 14 pts;
Line height: 16

Longer/Level 1 Subheds

Subhed 2

GT America Cond.
Bold, 12 pts, All caps;
Line height: 14

SHORTER/LEVEL 2 SUBHEDS

Annotation text 1

GT America Cond.
Reg., 12 pts;
Line height: 14

Annotation option 1 felis ut mollis dignissim, turpis purus ultricies libero, id bibendum ipsum nisl vitae metus

Annotation text 2

Same as 1 but 70% K
(#6D6E71)

Annotation option 2 felis ut mollis dignissim, turpis purus ultricies libero, id bibendum ipsum nisl vitae metus

Label text

GT America Cond.
Regular, 12 pts;
Line height: 14

Labels look
like this

Leader line

0.5 px wide, solid
dot arrowhead at
70% scale

Footnote text

GT America Cond.
Reg. Italic, 11 pts;
Line height: 13;
60% K (#808285)

**Footnote sodales, ex sit amet hendrerit placerat, metus lectus dapibus turpis, at tempor urna felis quis turpis.*

All text
Same as specs as
News Graphic Desktop →

Headline for Opinion Graphic on Desktop: Default Width

Intro text lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis. Aliquam convallis felis vitae metus tempus, non iaculis odio tristique.

Background
#E8EBEE

Longer/Level 1 Subheds

Annotation option 1 felis ut mollis dignissim, turpis purus ultricies libero, id bibendum ipsum nisl vitae metus

SHORTER/LEVEL 2 SUBHEDS

Annotation option 2 felis ut mollis dignissim, turpis purus ultricies libero, id bibendum ipsum nisl vitae metus

Labels look like this —————•

**Footnote sodales, ex sit amet hendrerit placerat, metus lectus dapibus turpis, at tempor urna felis quis turpis.*

Headline for Opiniom Graphic on Desktop: Text Width

Intro text lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis. Aliquam convallis felis vitae metus tempus, non iaculis odio tristique.

Headline for News Graphic on Desktop: Full Width

All text
Same specs as News
Graphic Mobile

Headline for Opinion Graphic on Mobile

Intro text lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis. Aliquam convallis felis vitae metus tempus, non iaculis odio tristique.

Longer/Level 1 Subheds Look Like This

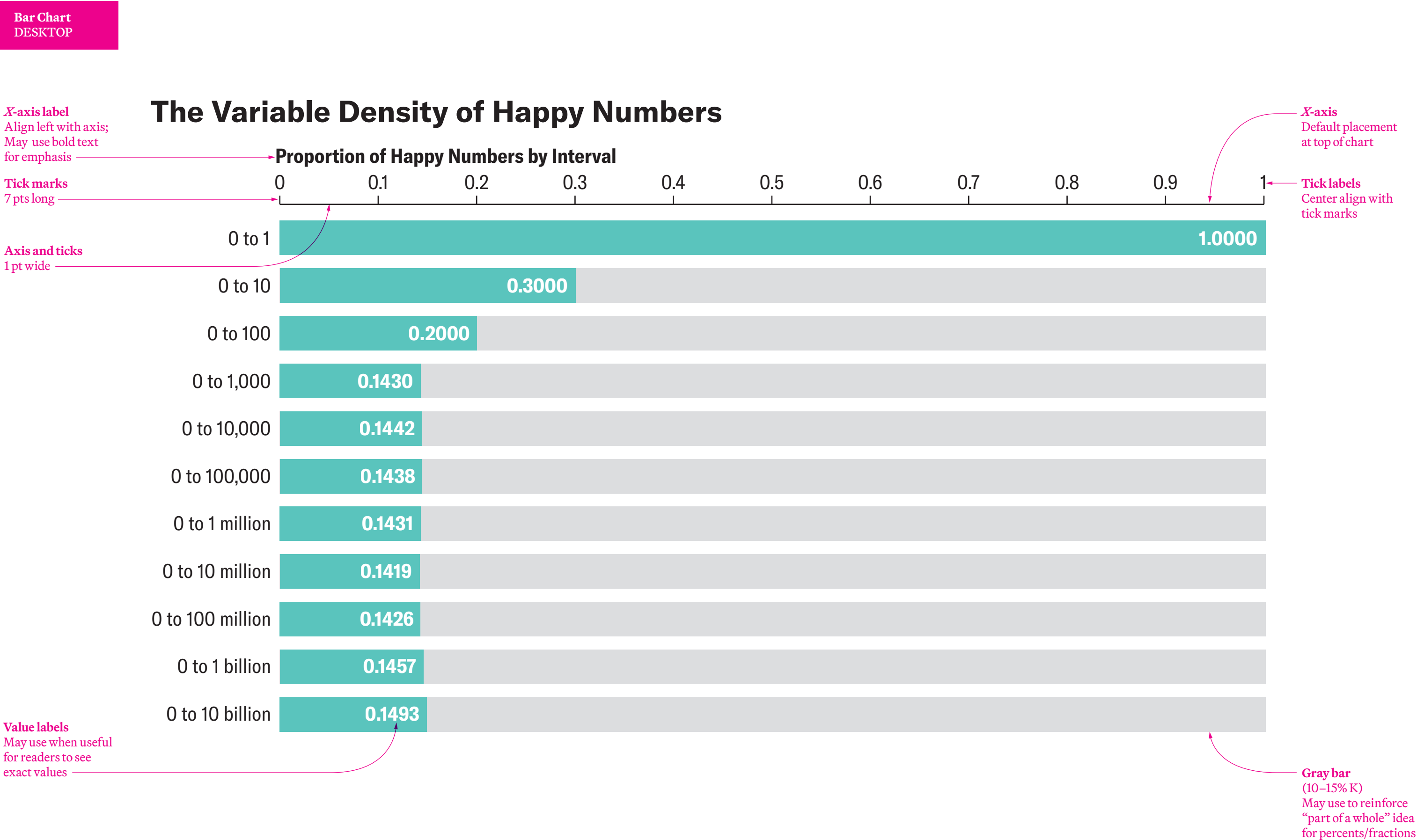
SHORTER/LEVEL 2 SUBHEDS

Annotation option 1 felis ut
mollis dignissim, turpis purus
ultricies libero, id bibendum
ipsum nisl vitae metus

Annotation option 2 felis ut
mollis dignissim, turpis purus
ultricies libero, id bibendum
ipsum nisl vitae metus

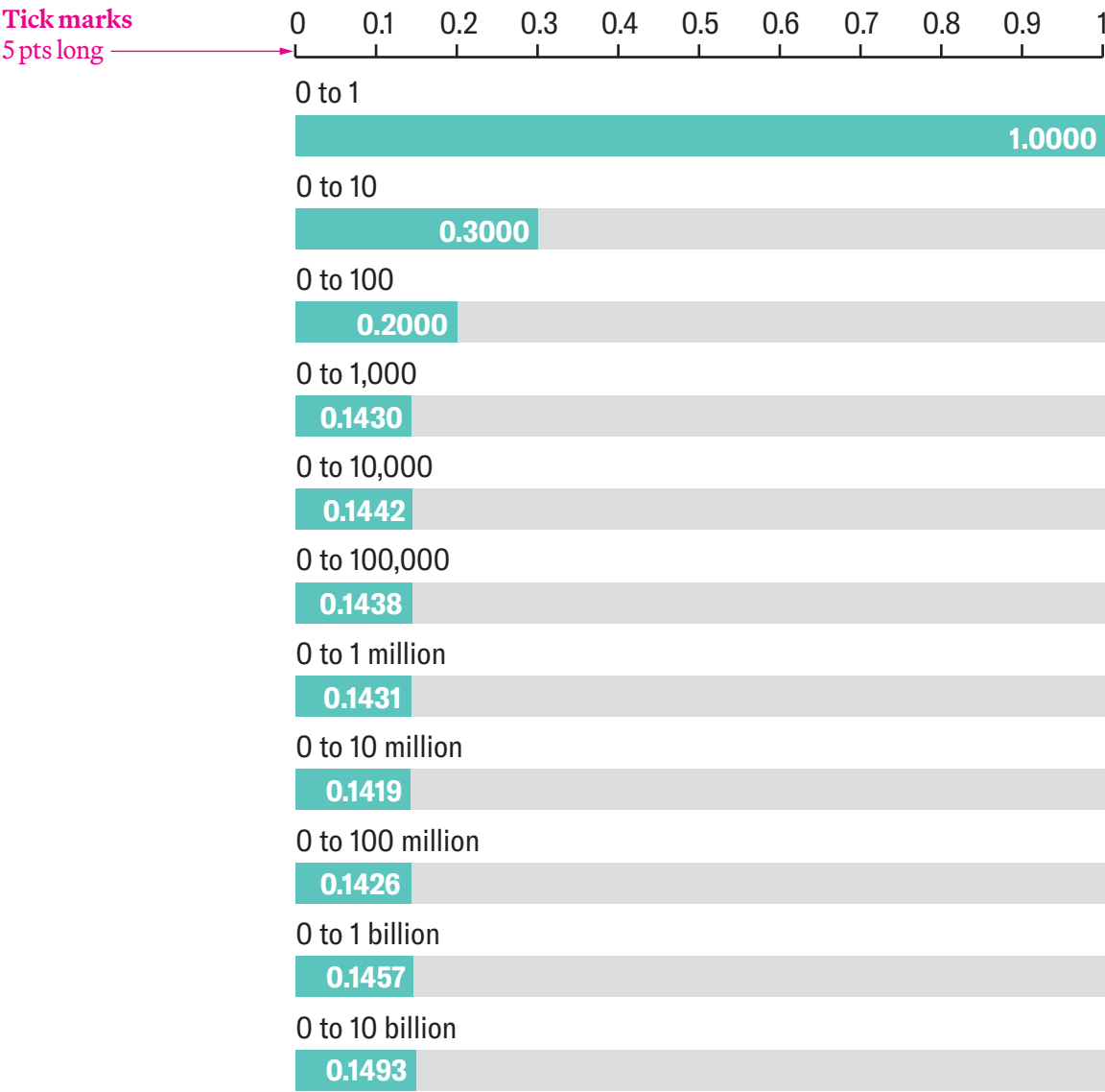
Labels look
like this —————•

**Footnote sodales, ex sit amet hendrerit placerat, metus lectus dapibus
turpis, at tempor urna felis quis turpis.*



The Variable Density of Happy Numbers

Proportion of Happy Numbers by Interval

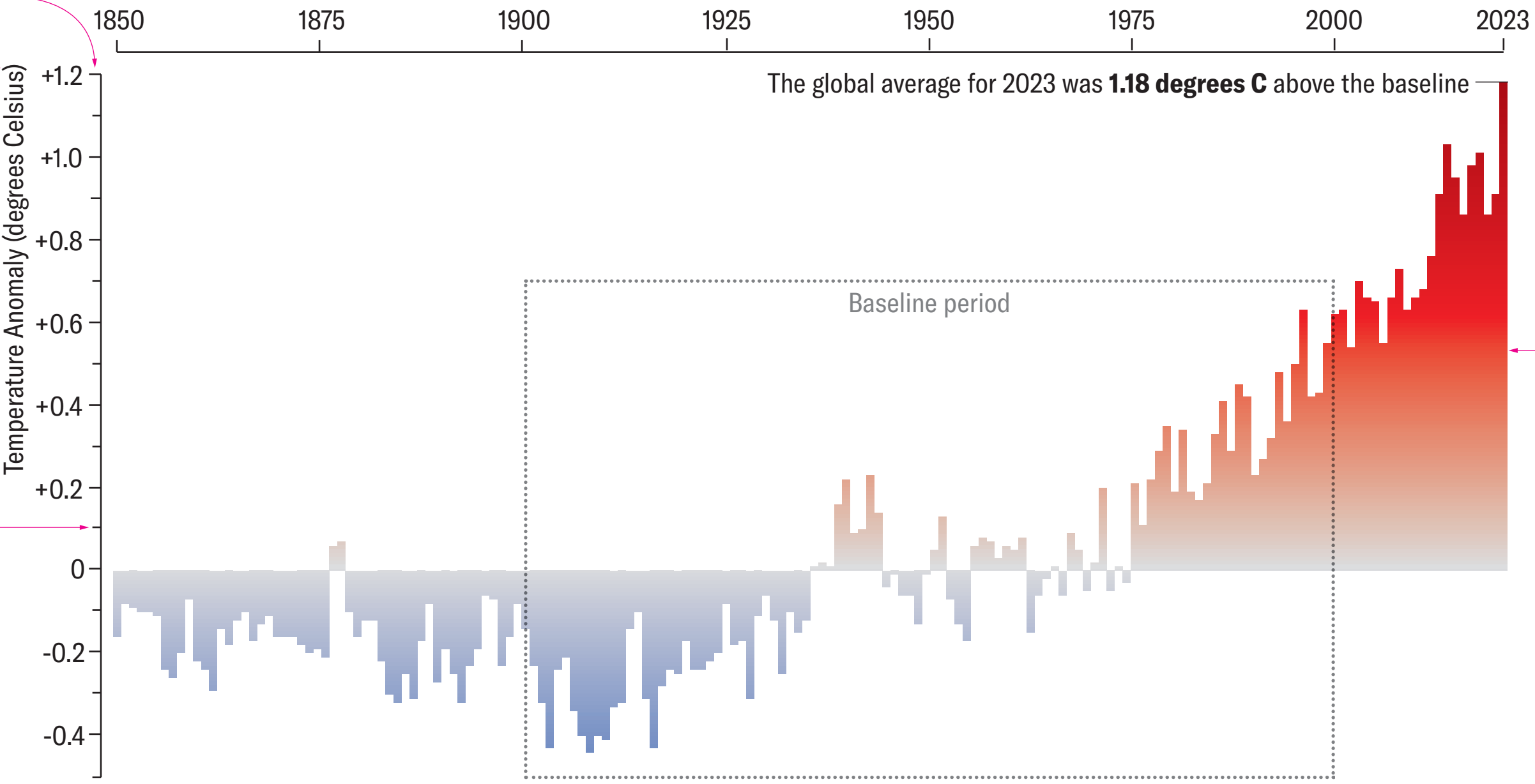


Y-axis
Default placement
on left side

Y-axis label
Align top with axis

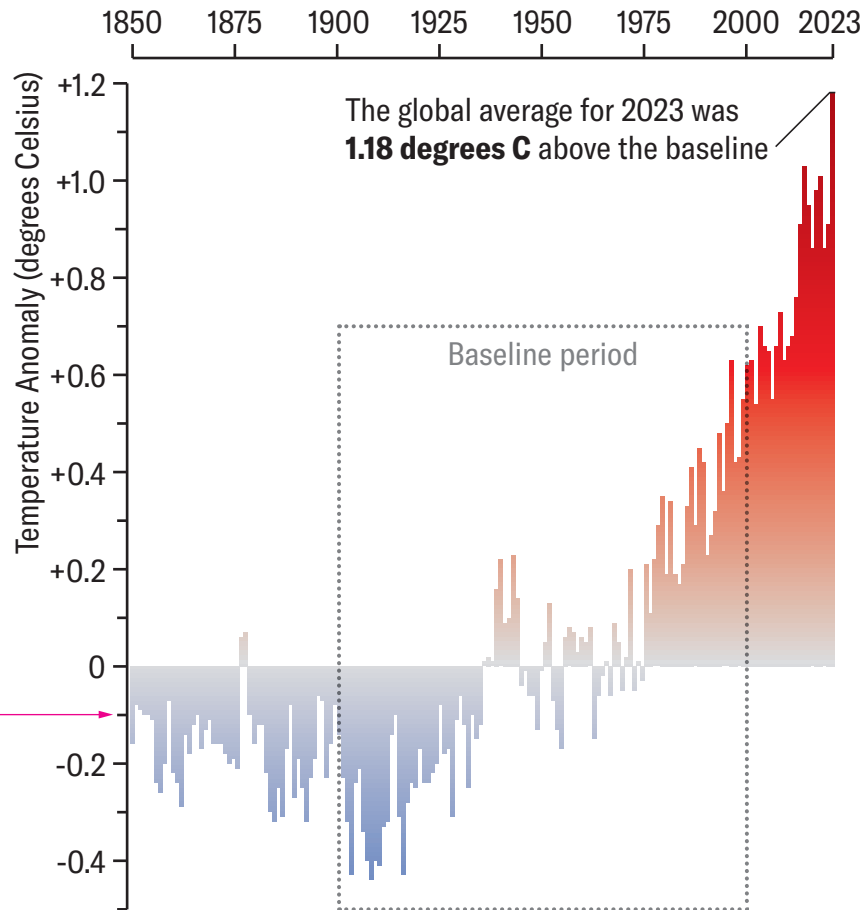
Unlabeled ticks
5 px long

Annual Global Temperature Anomalies, Compared with 1901–2000

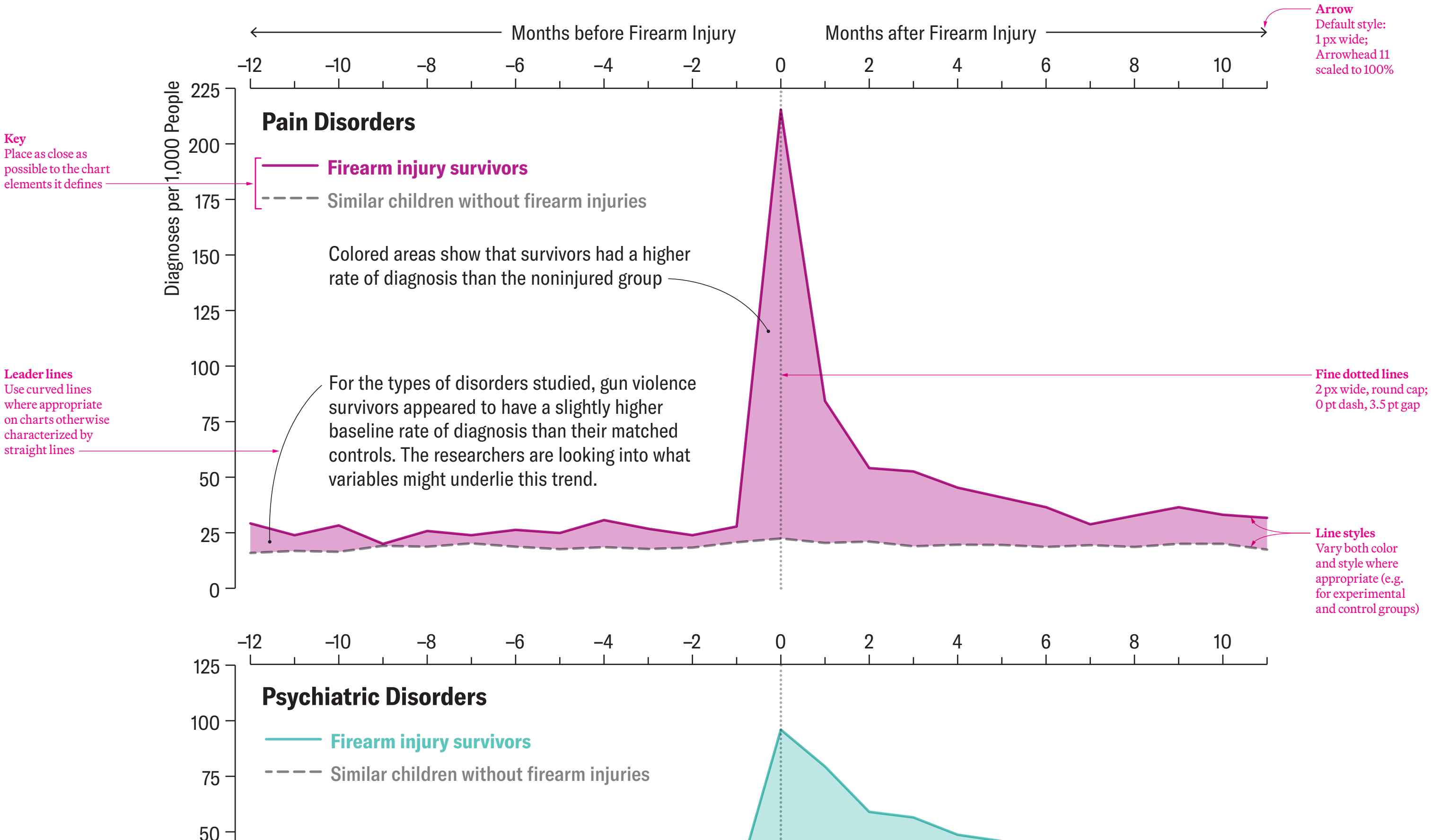


Color gradient
May be used to
emphasize
significance of
values

Annual Global Temperature Anomalies, Compared with 1901–2000



How Firearm Injuries Impact Children’s Health

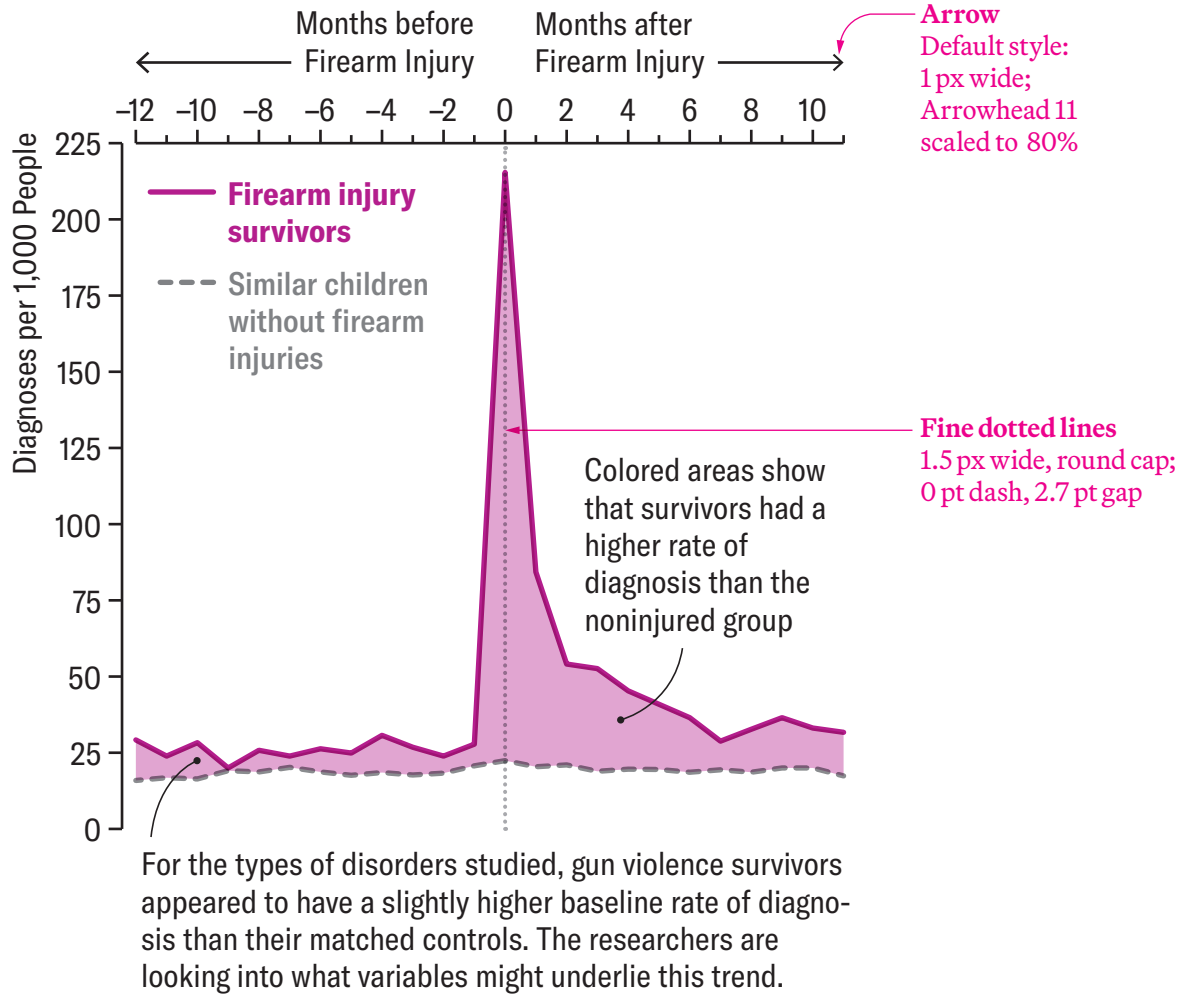


Headline

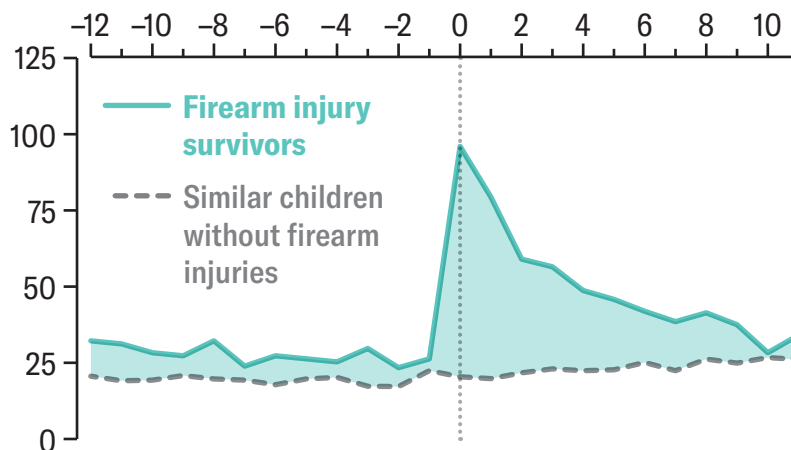
Adjust text wrapping
as needed to avoid
“widows”

How Firearm Injuries Impact Children’s Health

Pain Disorders



Psychiatric Disorders

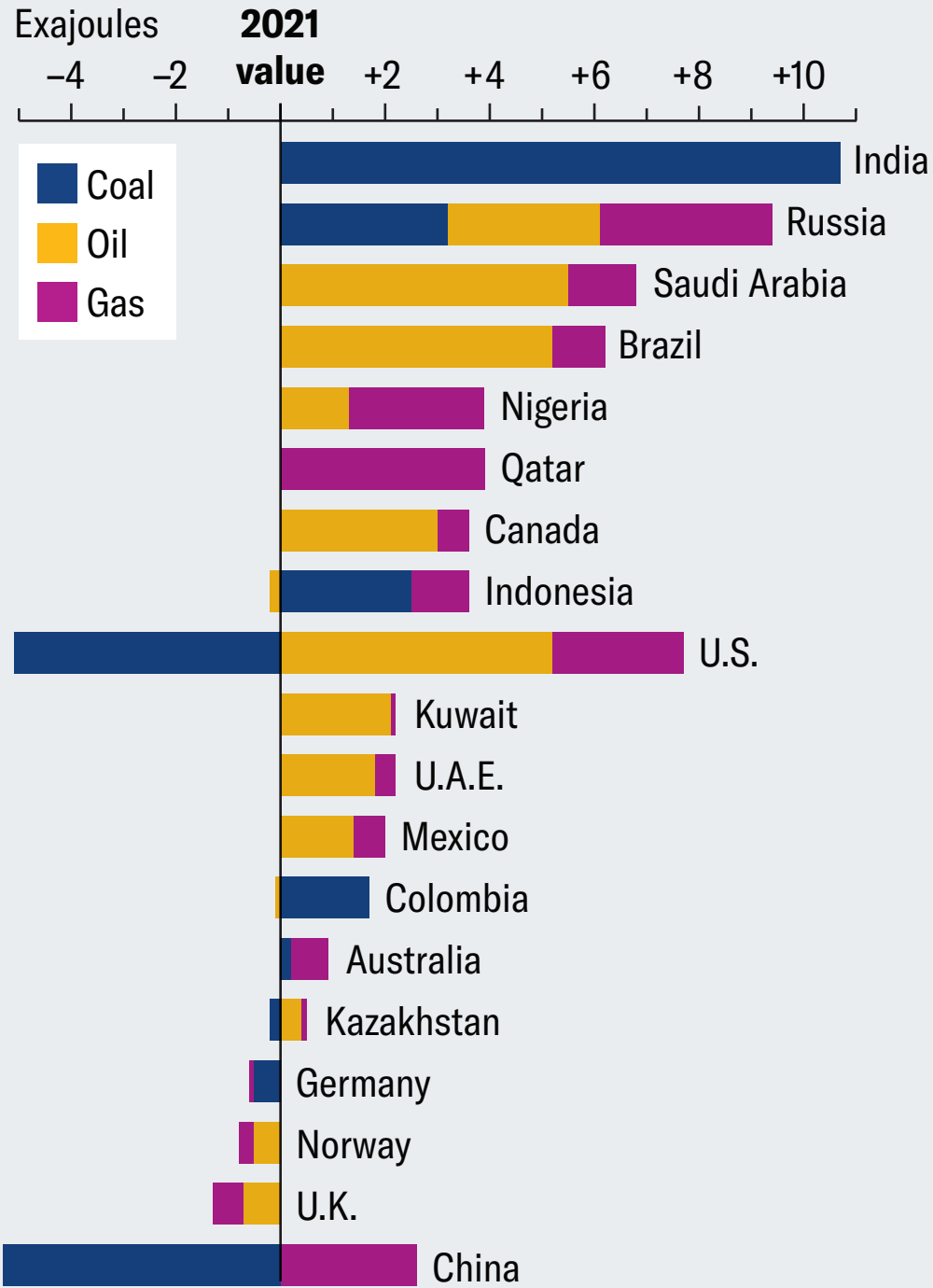


Substance Use Disorders

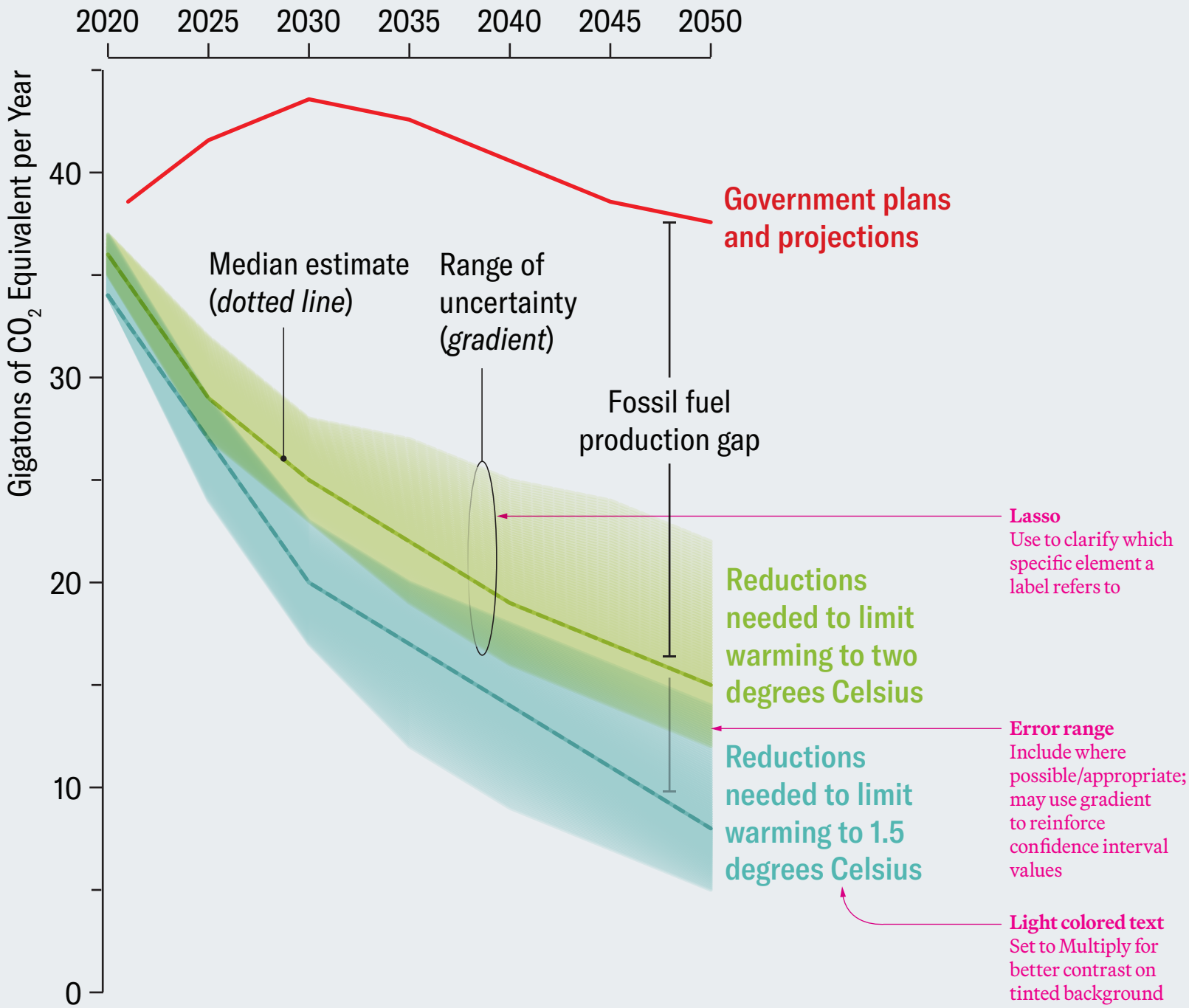


Chart headings
Use Subbhed 1 style
for multiple charts
on one graphic

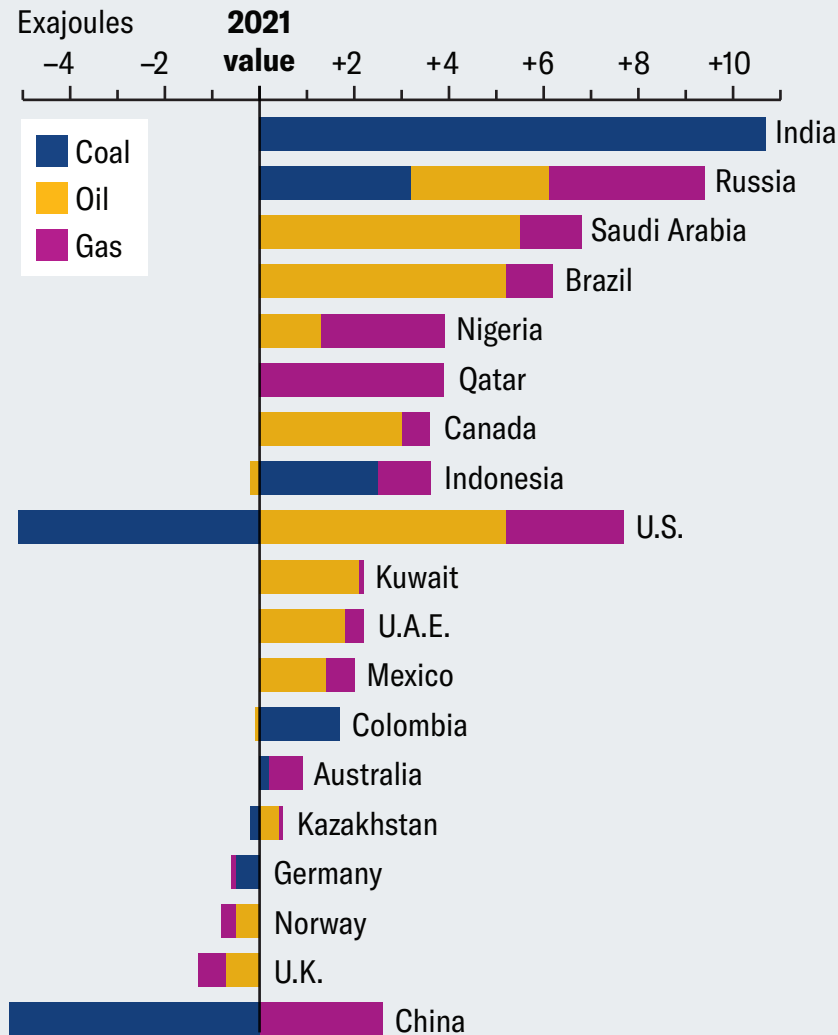
Countries' Planned Changes in Annual Fossil Fuel Production for 2030, Compared with 2021



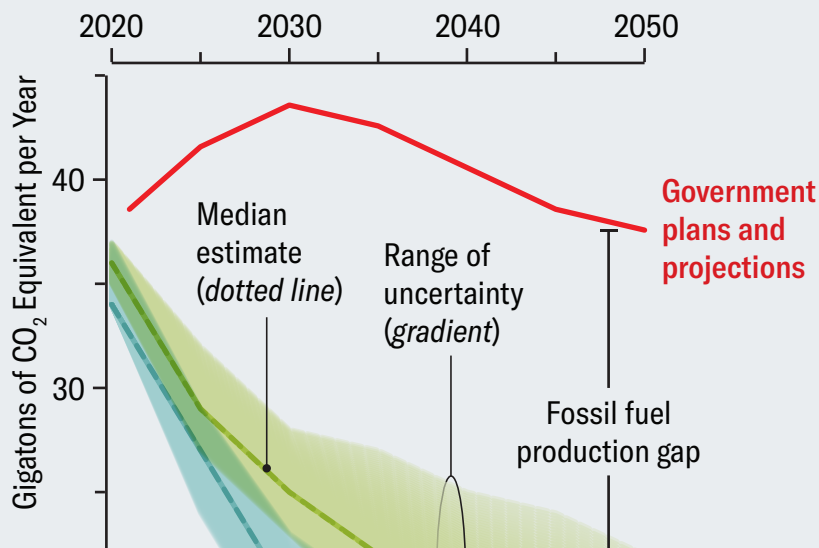
The “Production Gap” between Projected Global Greenhouse Gas Emissions and Climate Goal Scenarios



Countries' Planned Changes in Annual Fossil Fuel Production for 2030, Compared with 2021



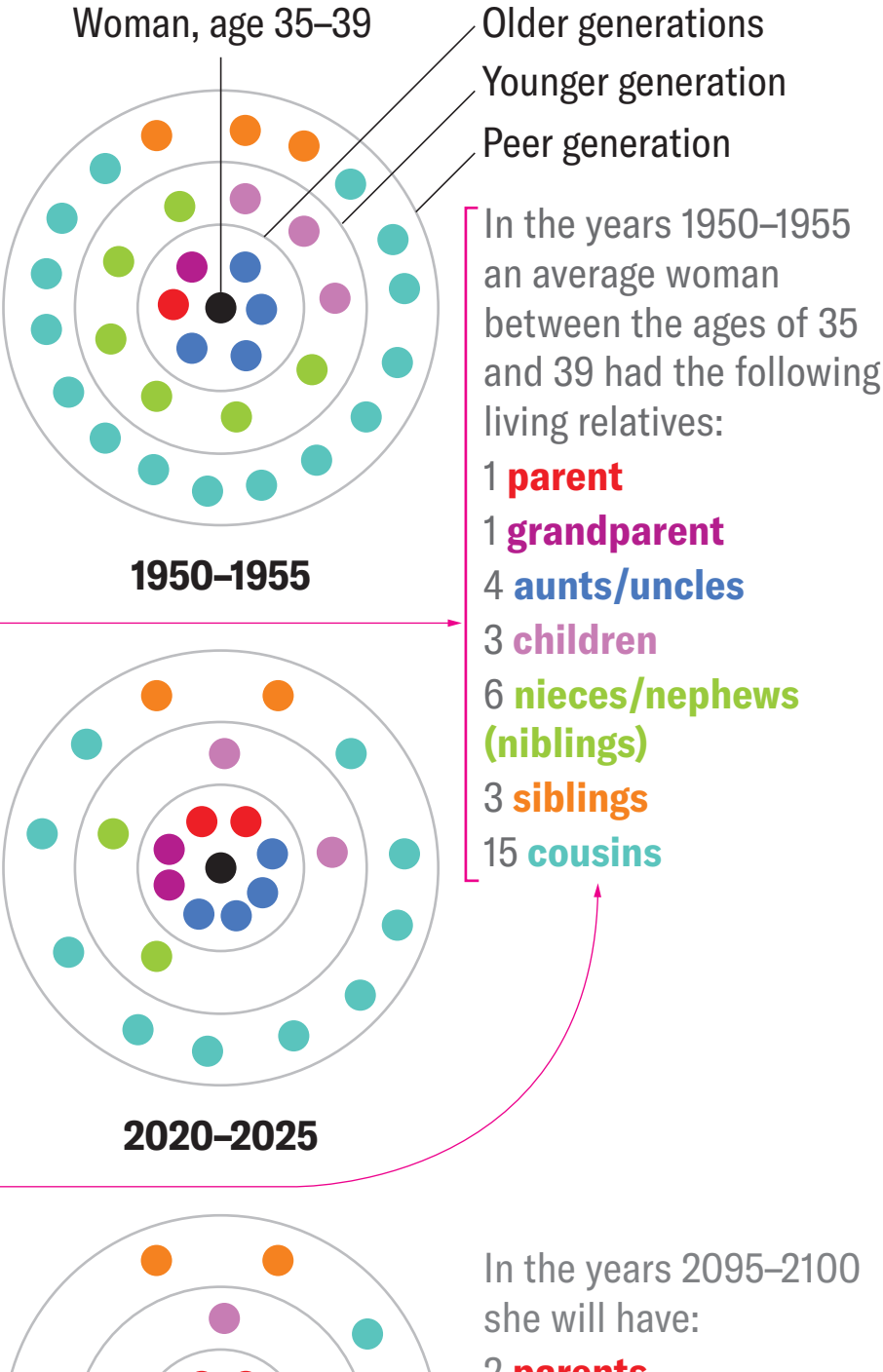
The "Production Gap" between Projected Global Greenhouse Gas Emissions and Climate Goal Scenarios



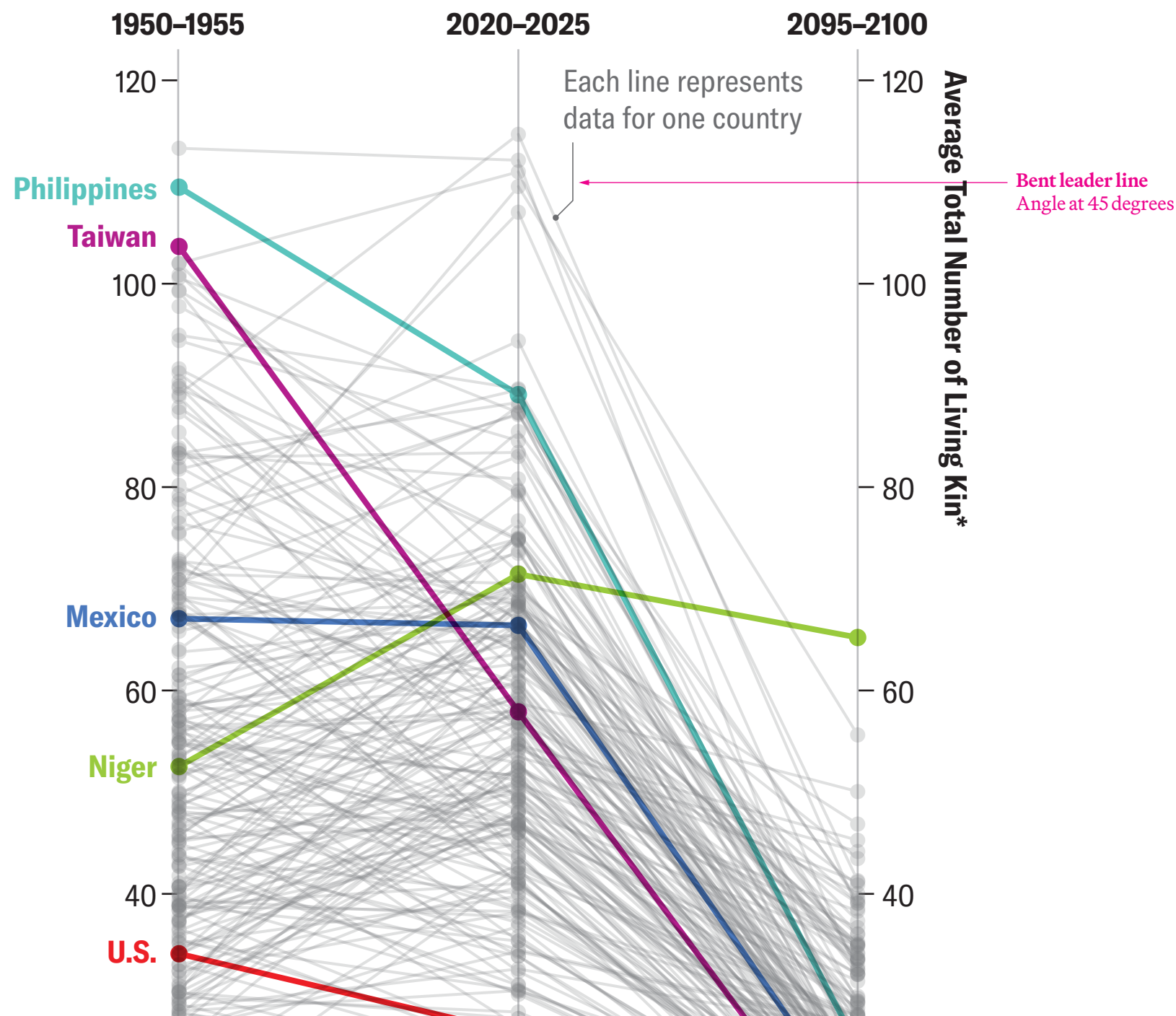
Changes in Kinship Networks Vary by Country

To analyze shifting family structures around the world, researchers focused on a randomly selected female individual in each country and estimated how many of each type of relative she would have based on her age during a specified interval of time. The periods 1950–1955, 2020–2025, and 2095–2100 reveal shifts over 150 years.

Typical Family Structure of a Woman in the U.S., Age 35–39



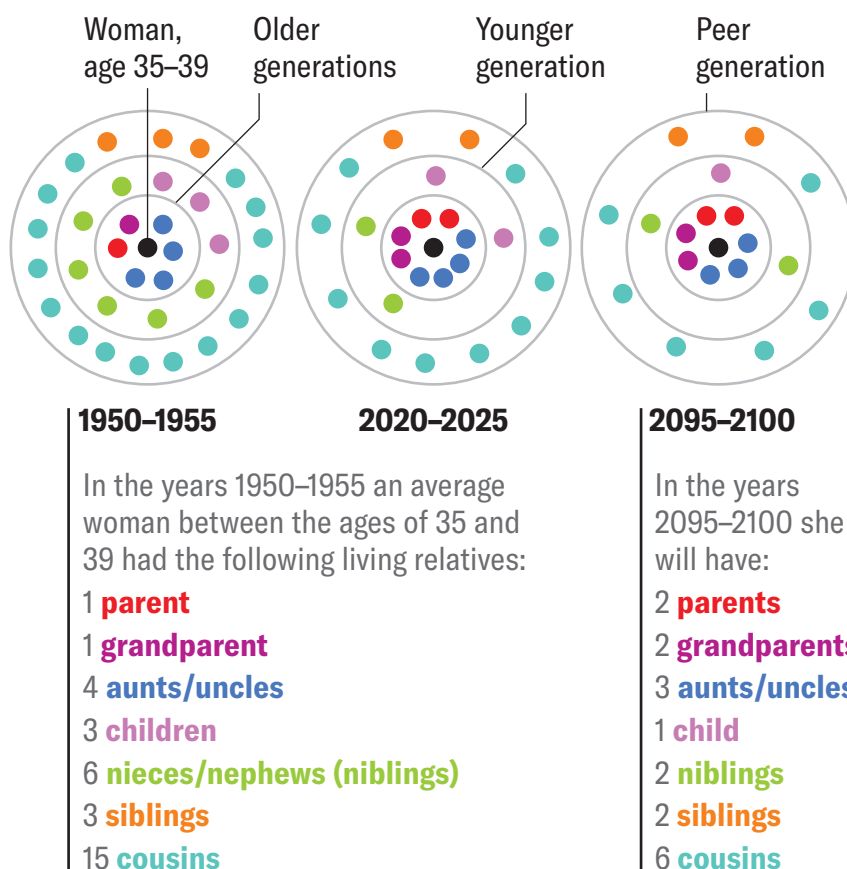
Estimated Number of Living Kin for a Woman, Age 65–69, by Country



Changes in Kinship Networks Vary by Country

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Estimated Number of Living Kin for a Woman, Age 65–69, by Country

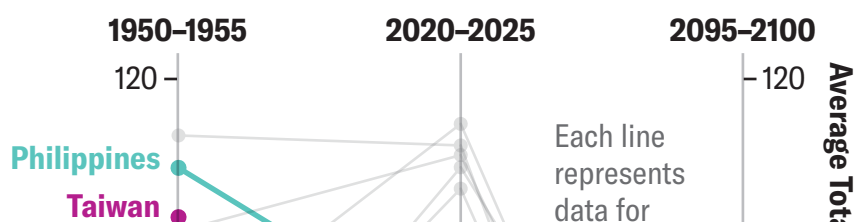
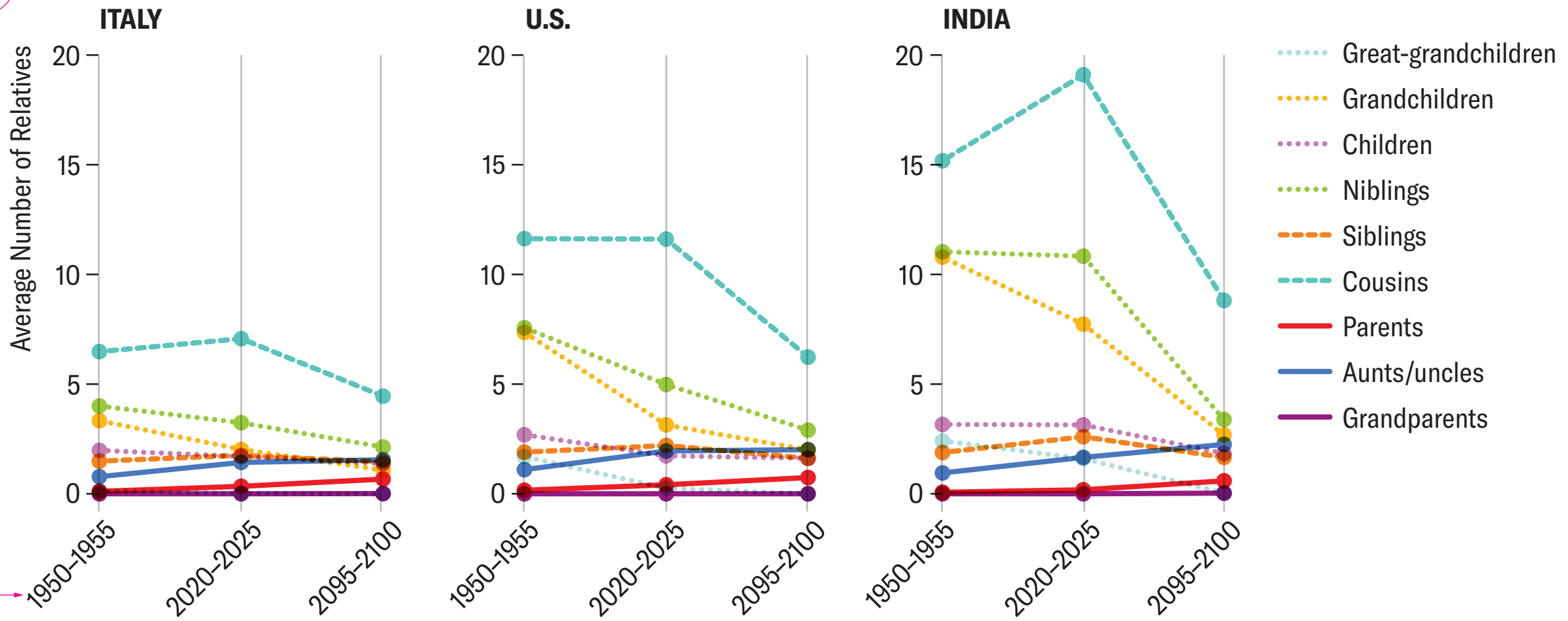
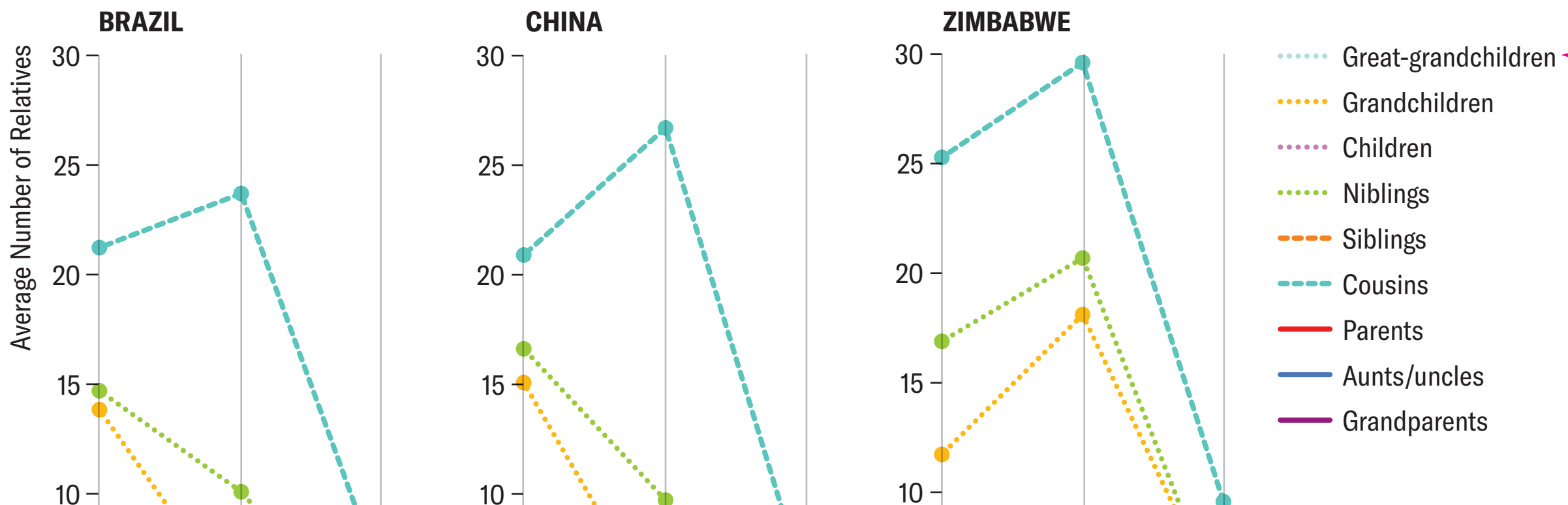


Chart headings
May use subhed style
for graphics that are
part of a series

Estimated Number of Living Kin for a Woman, Age 65–69, by Type of Relative

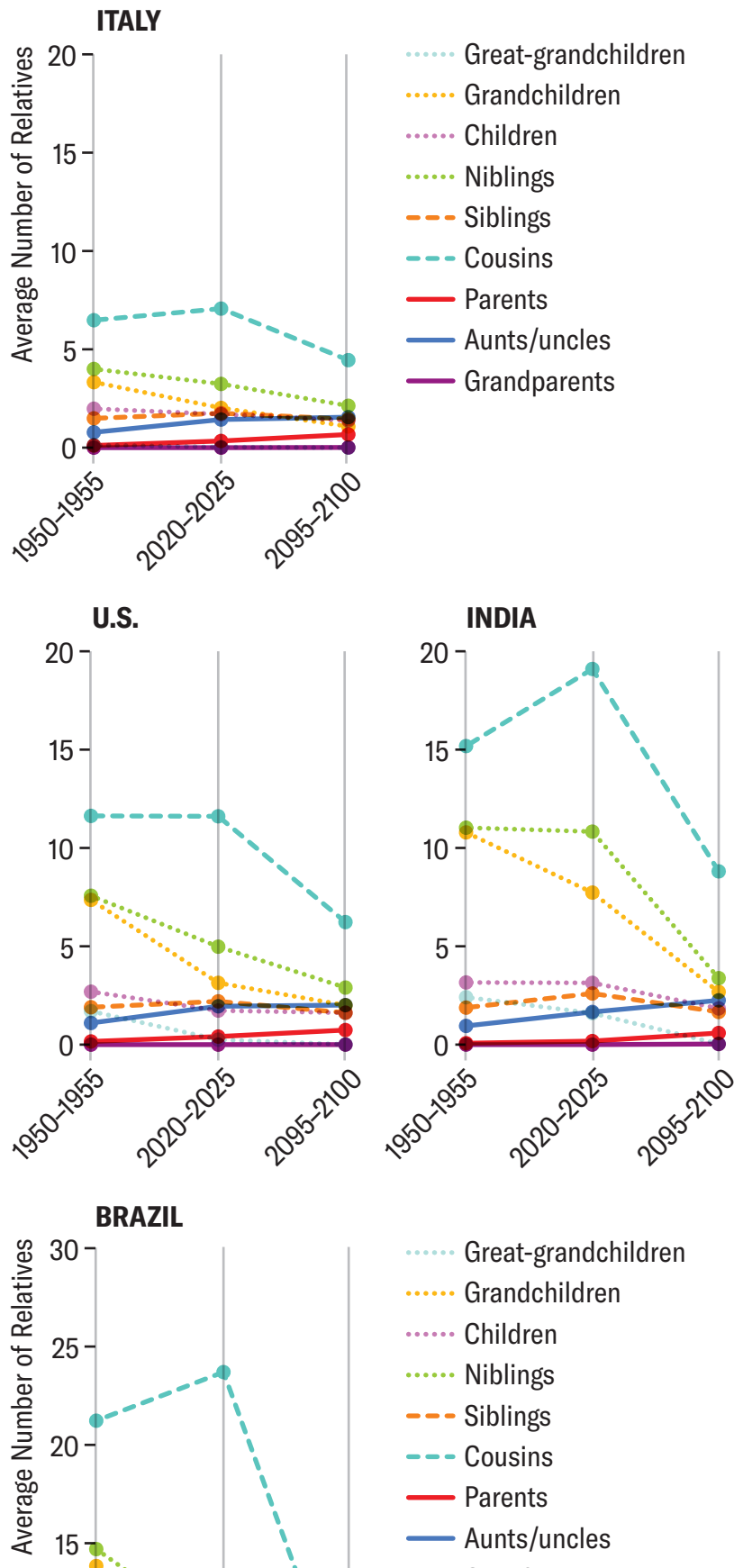


Labels
May rotate 45
degrees when
needed for fit



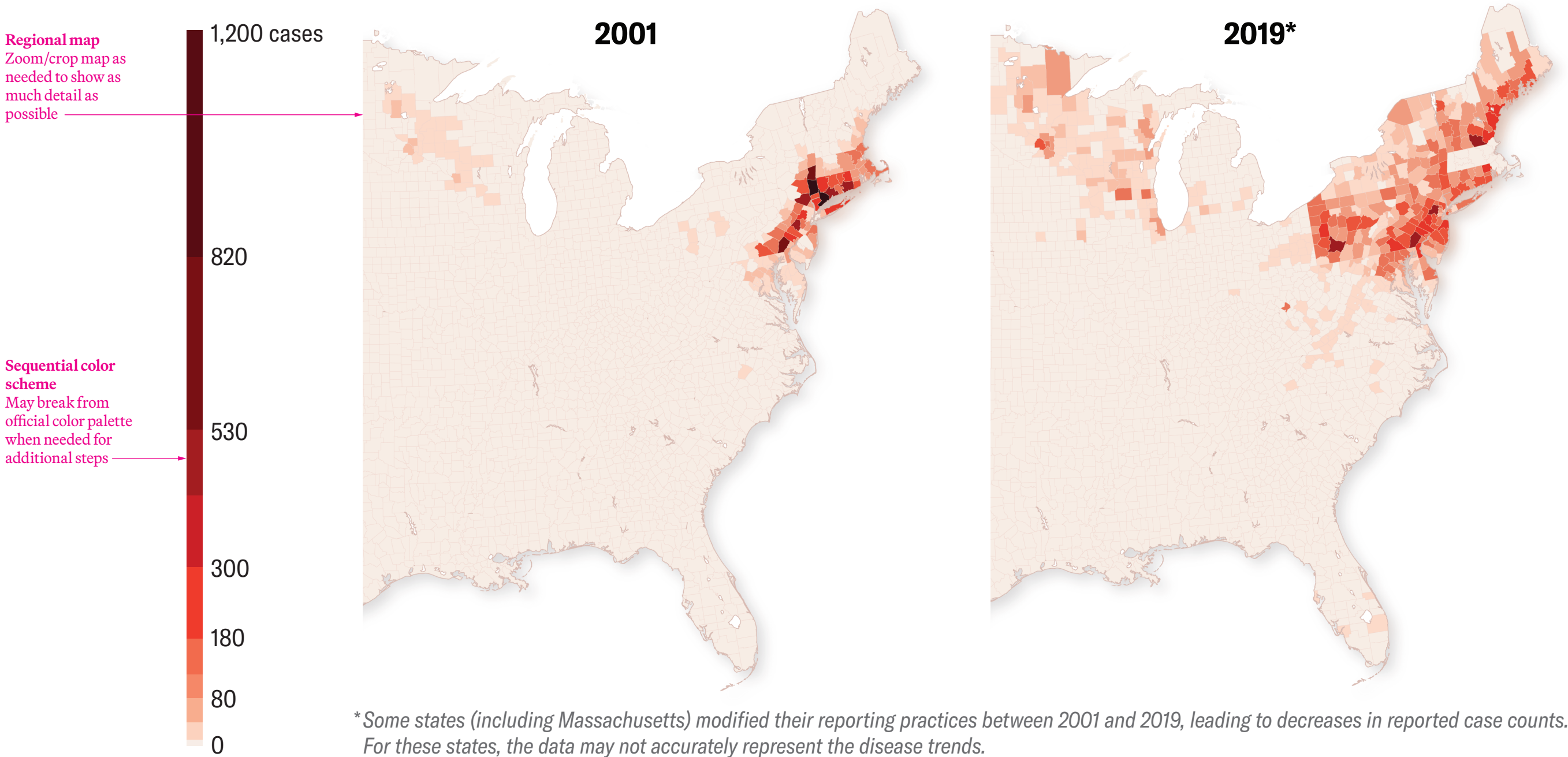
Key
May repeat as
needed to avoid
making readers
scroll back up

Estimated Number of Living Kin for a Woman, Age 65–69, by Type of Relative



Number of Reported Lyme Disease Cases by County

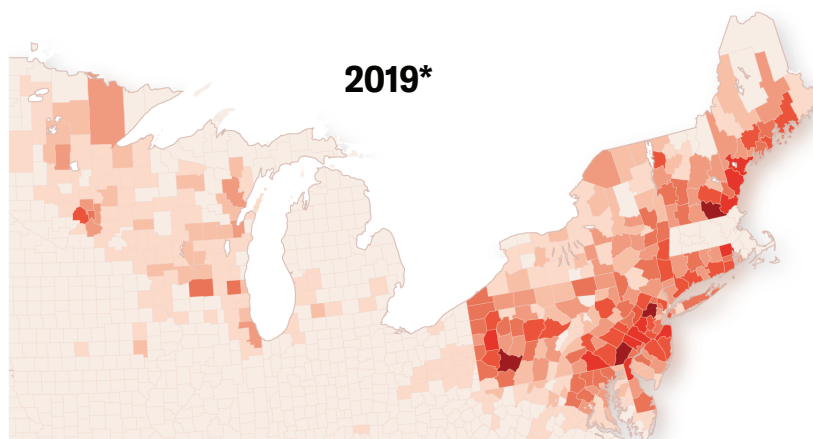
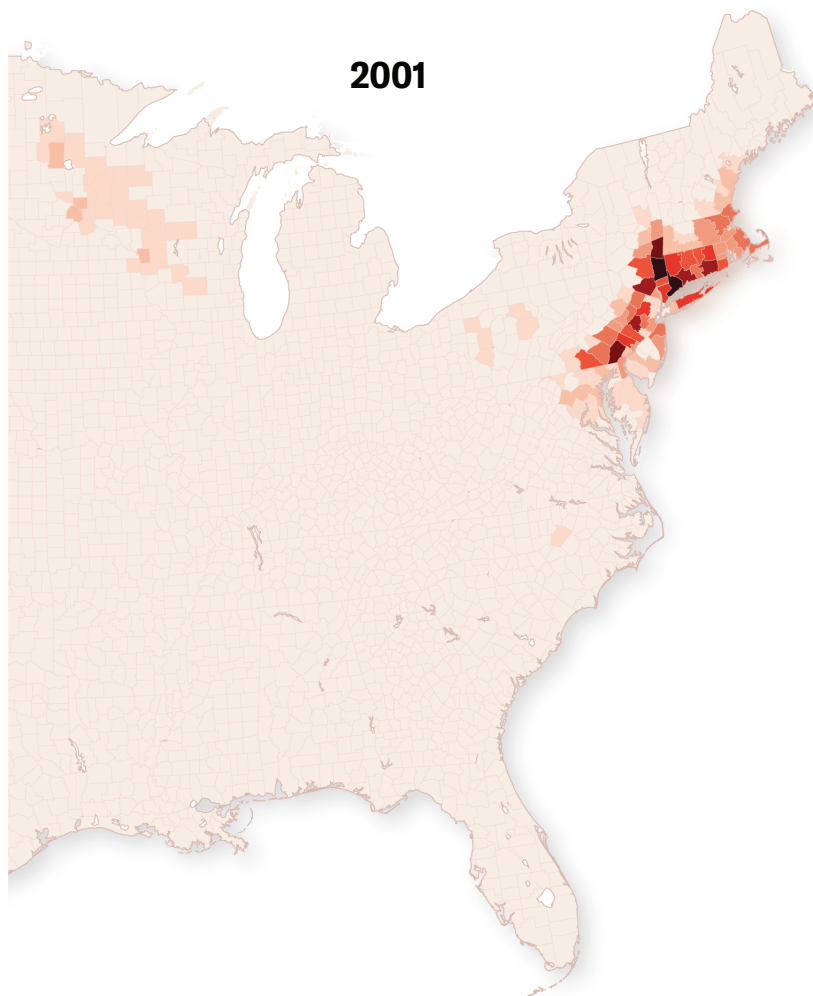
In 2019 more counties across the Northeast and the Midwest reported increased Lyme disease cases, compared with 2001.



**Some states (including Massachusetts) modified their reporting practices between 2001 and 2019, leading to decreases in reported case counts. For these states, the data may not accurately represent the disease trends.*

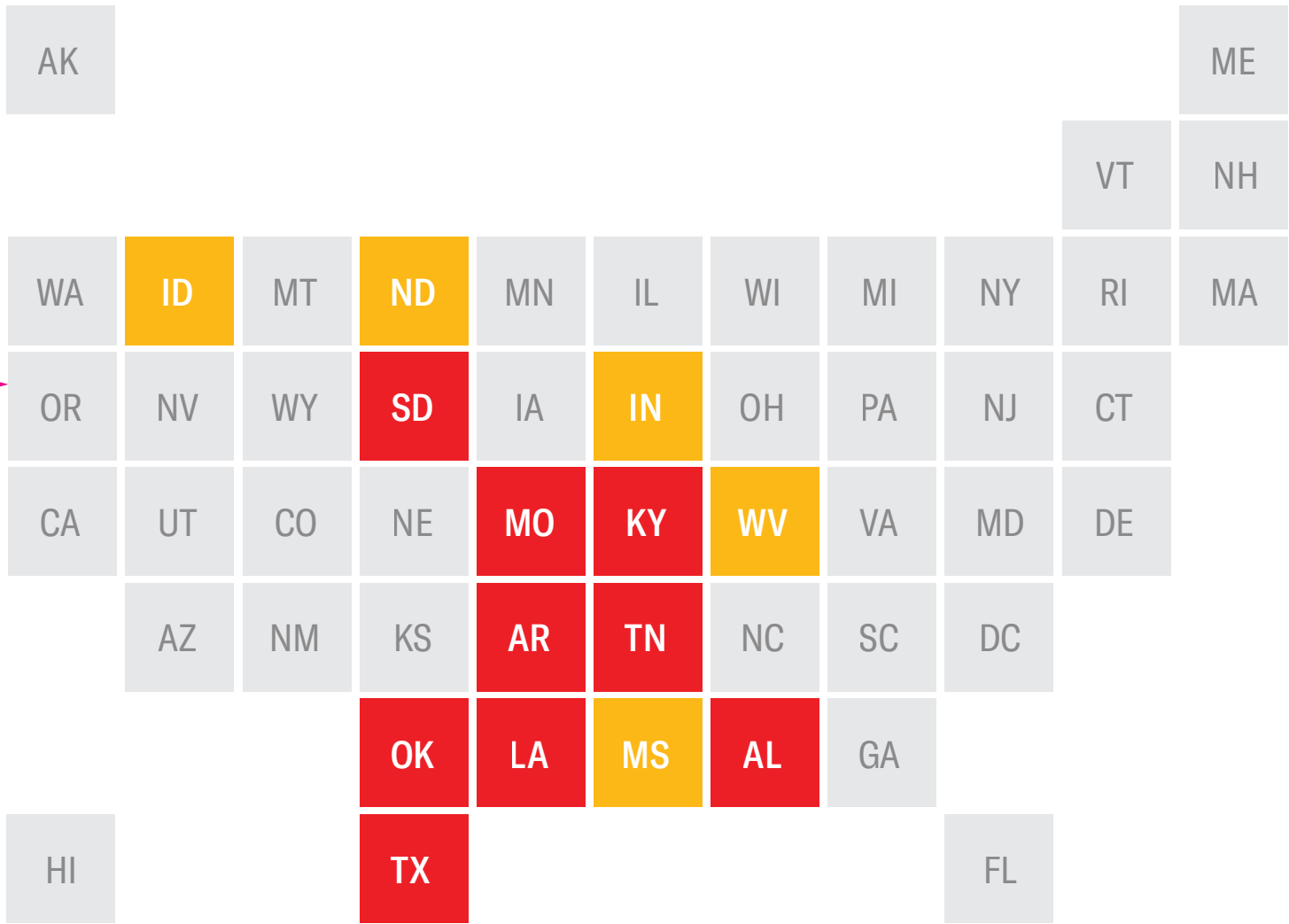
Number of Reported Lyme Disease Cases by County

In 2019 more counties across the Northeast and the Midwest reported increased Lyme disease cases, compared with 2001.



Which States Have Implemented a Total Abortion Ban since *Dobbs*?

Tile map
Use for generally familiar regions when geographical patterns are absent or irrelevant to the data



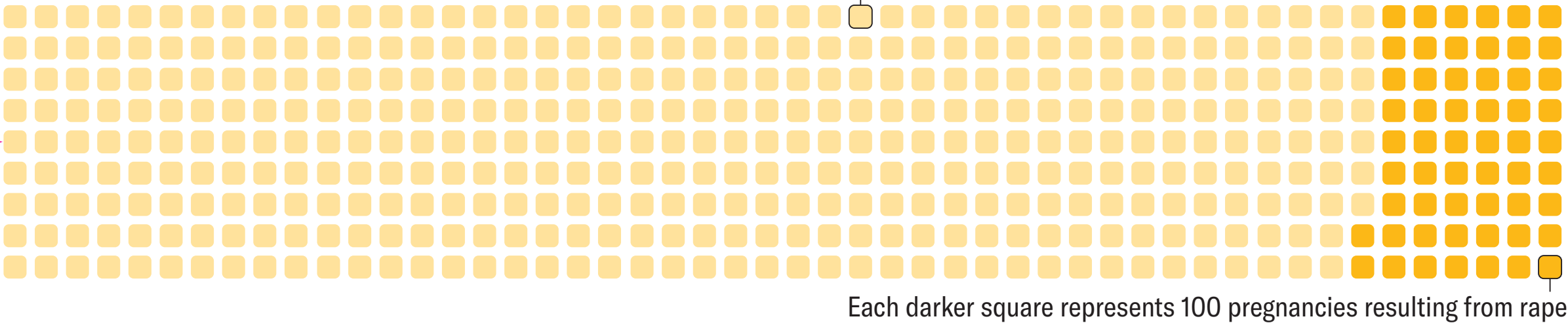
States that have banned abortion throughout pregnancy since the Supreme Court’s decision to overturn *Roe v. Wade* in *Dobbs v. Jackson Women’s Health Organization* are highlighted on the map.

- Ban includes an exception for rape (*yellow*)
- Ban has no exception for rape (*red*)

How Many Rapes Have Resulted in Pregnancy in Those States since Their Ban Went into Effect?

IN STATES WITH RAPE EXCEPTIONS

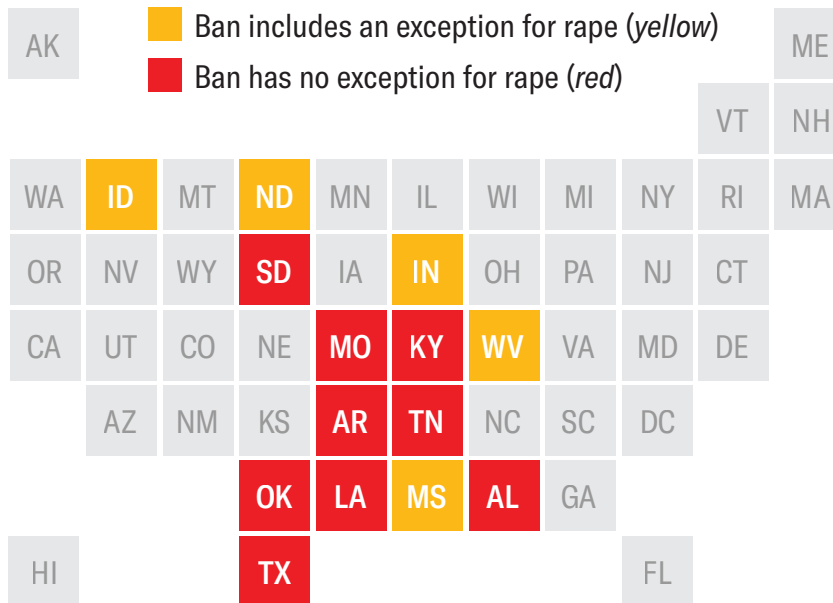
Icon array
Use to emphasize scale of values and/or to reinforce idea of data points as individual entities (i.e. humans)



On-art identifiers
Use in lieu of a separate key where appropriate

Which States Have Implemented a Total Abortion Ban since *Dobbs*?

States that have banned abortion throughout pregnancy since the Supreme Court's decision to overturn *Roe v. Wade* in *Dobbs v. Jackson Women's Health Organization* are highlighted on the map.

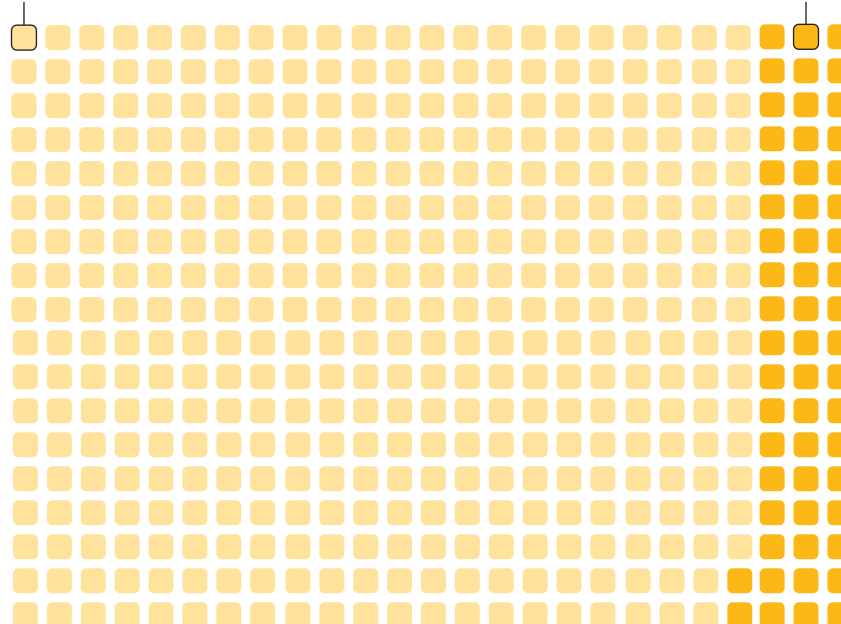


How Many Rapes Have Resulted in Pregnancy in Those States since Their Ban Went into Effect?

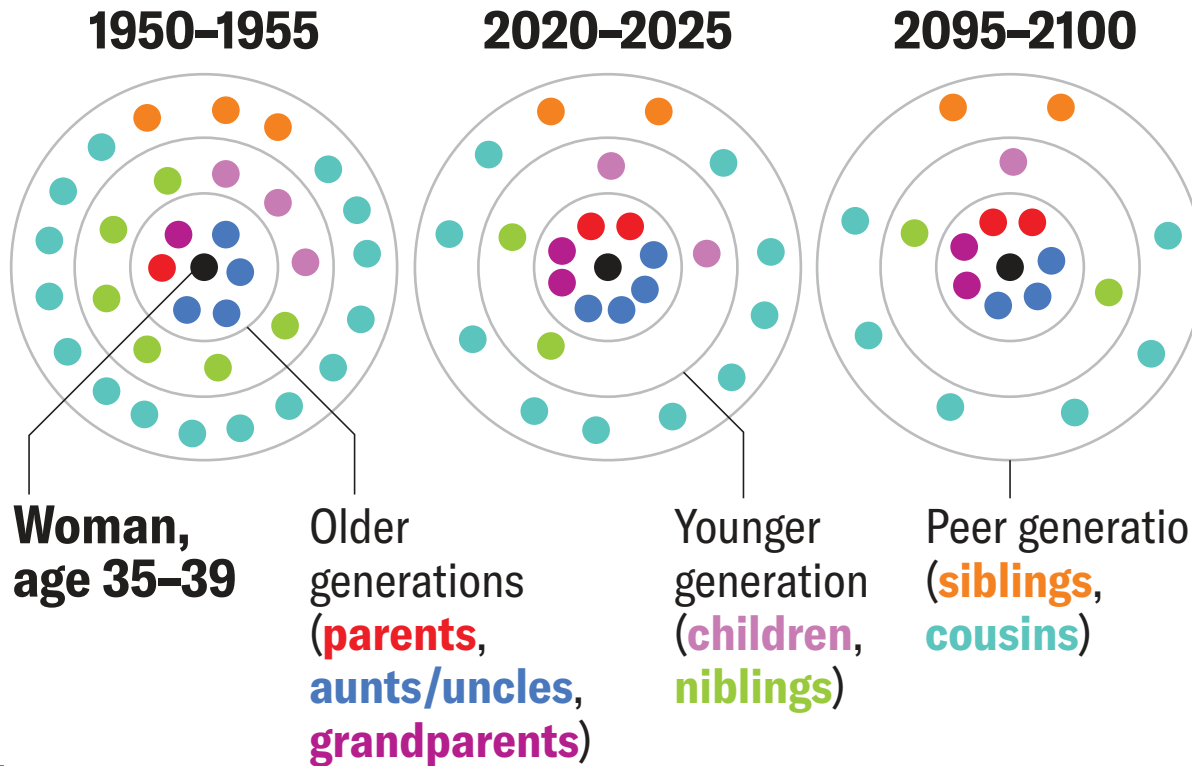
IN STATES WITH RAPE EXCEPTIONS

Each square represents
100 instances of vaginal rape

Each darker square represents
100 pregnancies resulting from rape



Typical Family Structure of a Woman in the U.S., Age 35–39



Graphic width
500 px

Heading
Align left; default
size 21–24 pts; scale
to fit well on one line
(or to avoid widows)

Margin
~20 pts all around

Graphic should be
relatively small and
self-contained;
include minimum
amount of text
needed for comp-
rehension; intro or
other long text blocks
should be incorporated
into preceding text in
newsletter or associated
caption on social media

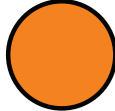
Labels/annotations
Default size: 18 pts

COLOR PALETTES

PURE COLOR HUES



CMYK 0.100.100.0
HEX #ED1C24
4.79 : 1
AA



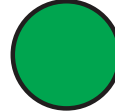
CMYK 0.60.100.0
HEX #F58220
8.1 : 1
AAA



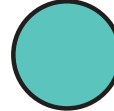
CMYK 0.30.100.0
HEX #FDB913
12.13 : 1
AAA



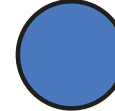
CMYK 45.0.100.0
HEX #9CCF0C
11.34 : 1
AAA



CMYK 100.0.100.0
HEX #00A651
6.57 : 1
AA



CMYK 60.0.30.0
HEX #39DBC81
12.14 : 1
AAA



CMYK 75.50.0.0
HEX #317FEE
5.4 : 1
AA

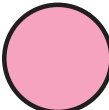


CMYK 65.90.0.0
HEX #763F98
7.16 : 1
AAA

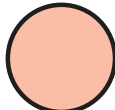


CMYK 30.100.0.0
HEX #B41E8E
5.97 : 1
AA

PURE COLOR + WHITE TINTS



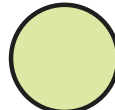
CMYK 0.45.5.0
HEX #F6A3BE
10.9 : 1
AAA



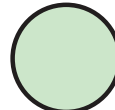
CMYK 0.30.30.0
HEX #FBBEA7
13.03 : 1
AAA



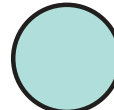
CMYK 0.10.45.0
HEX #FEE39C
16.72 : 1
AAA



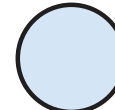
CMYK 15.0.45.0
HEX #DDFDA4
18.66 : 1
AAA



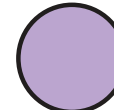
CMYK 20.0.25.0
HEX #CDE7CA
15.88 : 1
AAA



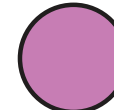
CMYK 30.0.15.0
HEX #B0DFDB
14.42 : 1
AAA



CMYK 15.5.0.0
HEX #D4E3F5
16.11 : 1
AAA



CMYK 25.35.0.0
HEX #C1A7EF
10.04 : 1
AAA



CMYK 20.60.0.0
HEX #E277DB
7.9 : 1
AAA

PURE COLOR + BLACK SHADES



CMYK 0.100.100.35
HEX #A70E13
7.73 : 1
AAA



CMYK 0.75.100.25
HEX #BE4F18
4.86 : 1
AA



CMYK 0.40.100.40
HEX #9C6D08
4.56 : 1
AA



CMYK 30.0.75.50
HEX #657F34
4.53 : 1
AA



CMYK 55.0.100.70
HEX #285511
8.71 : 1
AAA



CMYK 70.0.50.40
HEX #1D7F6A
4.89 : 1
AA



CMYK 100.75.0.25
HEX #004185
10.02 : 1
AAA



CMYK 75.100.0.35
HEX #481368
13.39 : 1
AAA

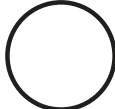


CMYK 40.100.0.10
HEX #951B82
13.39 : 1
AAA

BLACK, WHITE



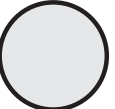
CMYK 0.0.0.100
HEX #000000
21 : 1
AA



CMYK 0.0.0.0
HEX #FFFFFF
21 : 1
AAA



CMYK 0.0.0.60
HEX #808080
5.32 : 1
AA



CMYK 0.0.0.10
HEX #E6E7E8
16.96 : 1
AAA